

ICN-Enabled Secure Edge Networking with Augmented Reality (ICE-AR)

Project Overview and Lessons Learned

NDN Community Meeting
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ICE-AR

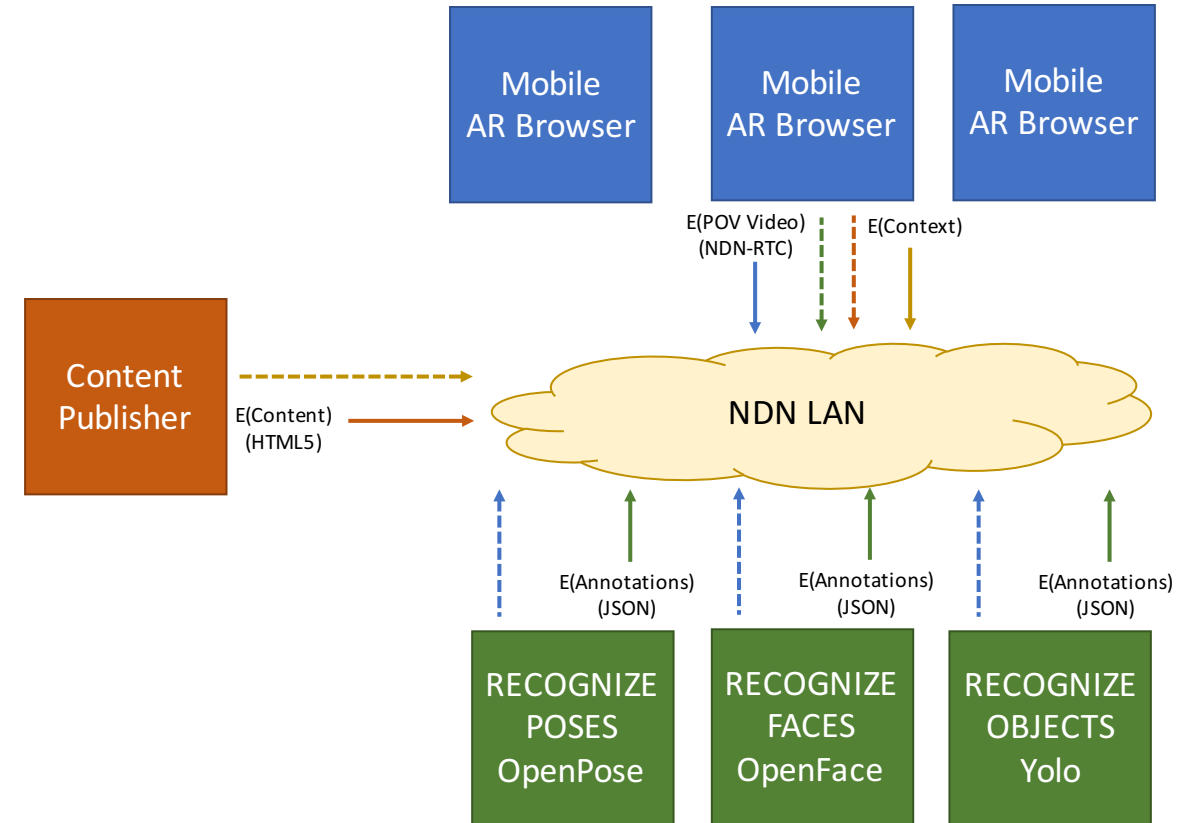
Using AR (and later MR) to drive research on NDN for the wireless edge.

- Applications, API, and library development
- Security at the edge
- Information-centric Wireless
- FPGA acceleration as a service

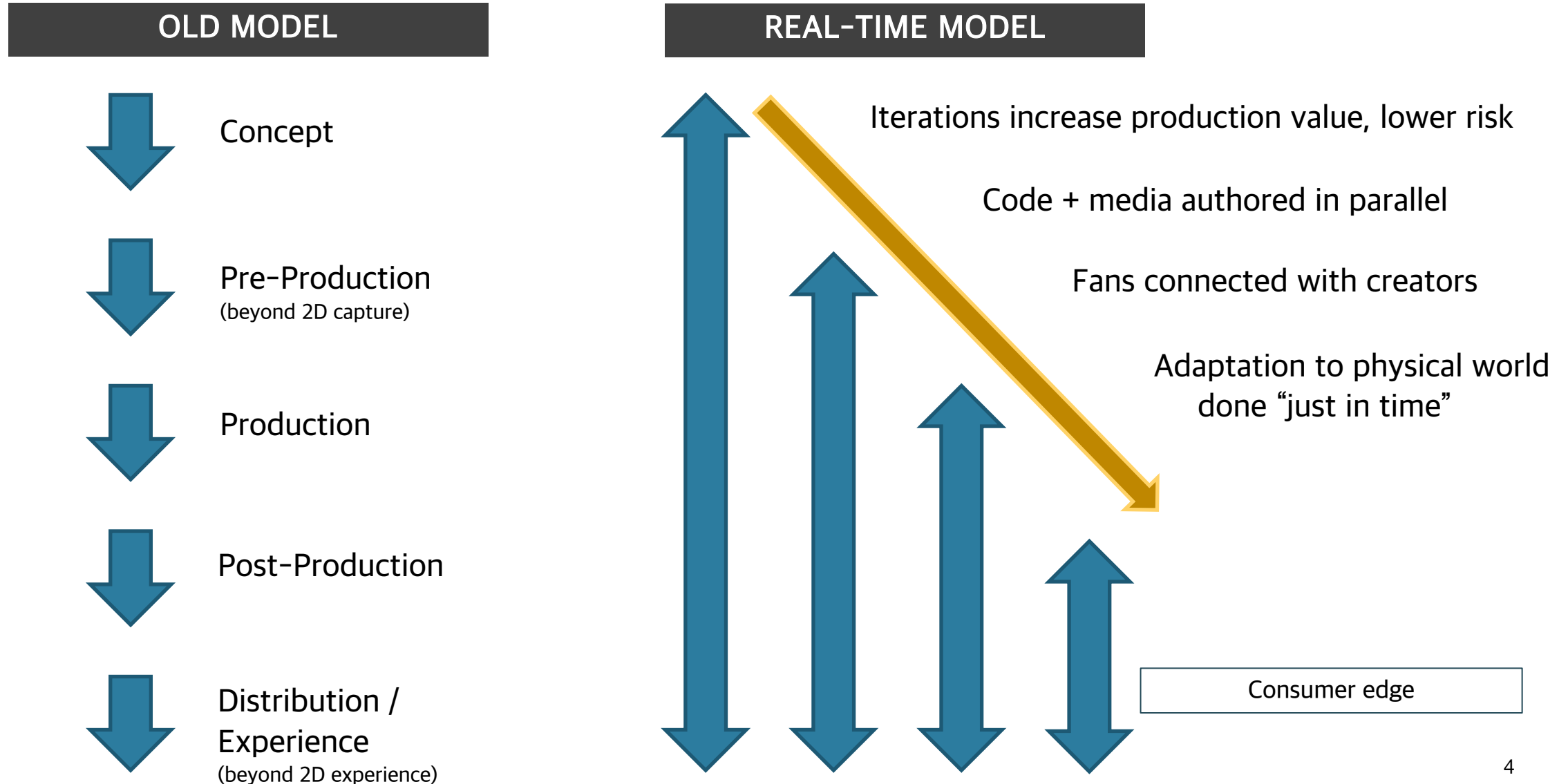
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Original Driver: Decentralized AR

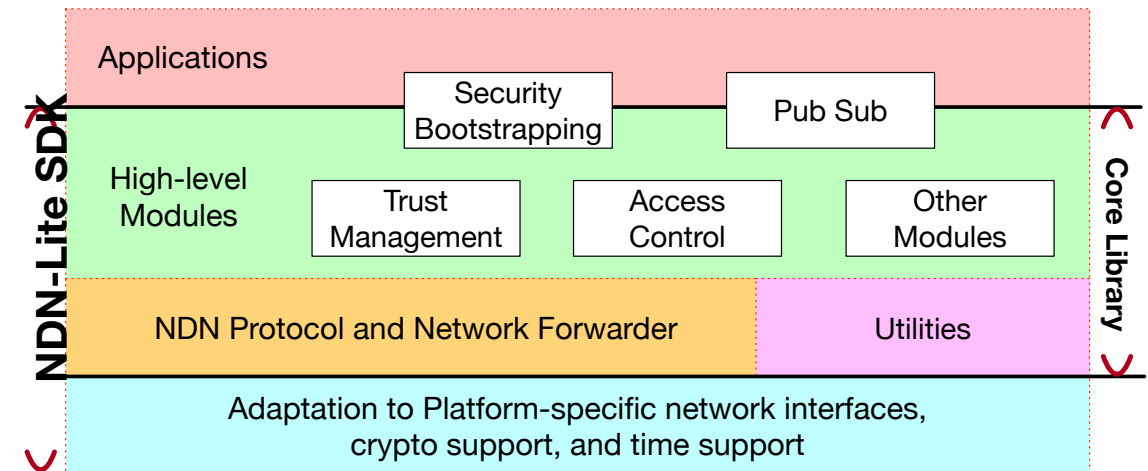
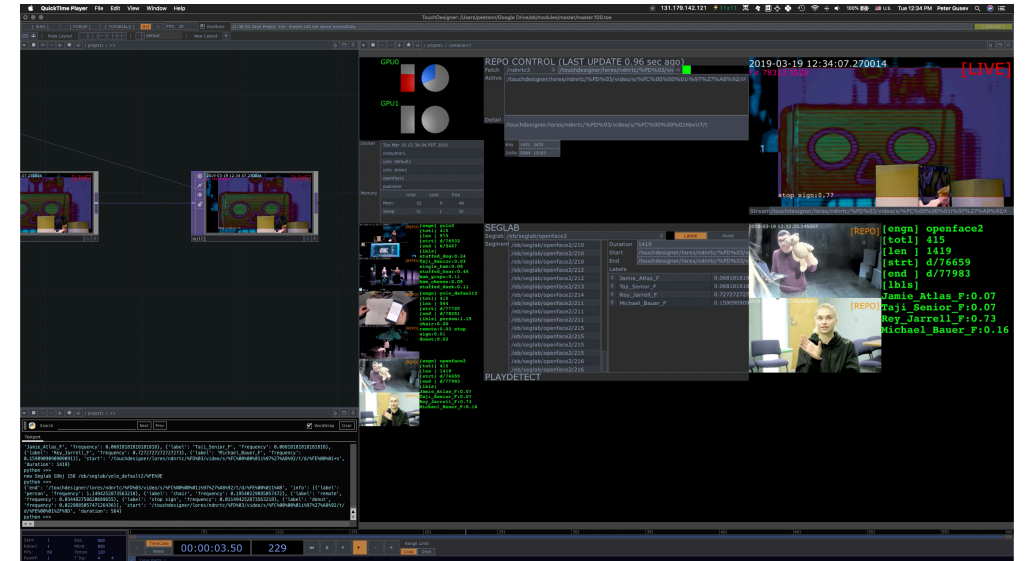


General Shift in Media Experience / Delivery



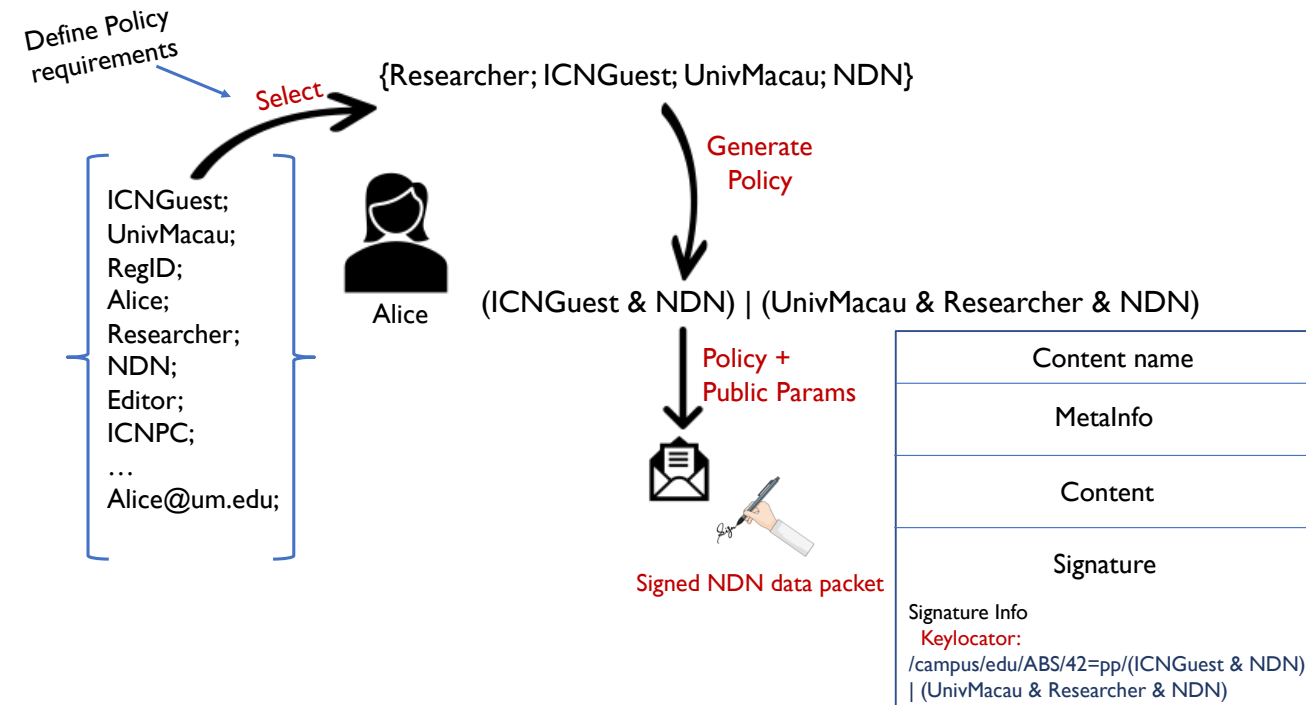
Applications, API, and library development

- ◇ Mobile AR, MR Integration Tests.
- ◇ Motivating new APIs / Libraries
 - **Common Name Library** (NDN-CNL) - Collection-oriented API
 - **TouchNDN** – Visual programming for data-centric media
 - **NDN-Lite** – IoT framework
 - **NDN RPS** – Approach for remote procedure calls
- ◇ Demonstration of integration with AOM AV1 video codec for consumer-driven spatial and temporal selectivity using names.



Security at the edge

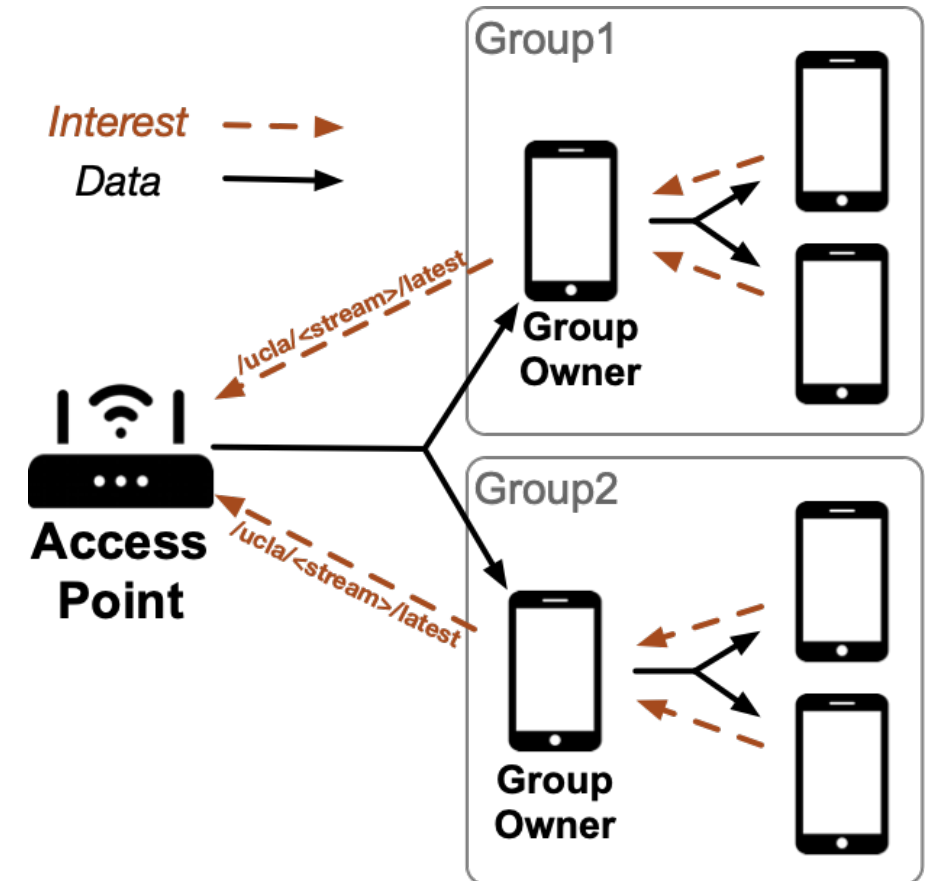
- ◇ Schematization of security relationship through names.
- ◇ **Usability**
 - NDNCERT development
 - Location-based certs
 - New signature (ABS) and certificate schemes (Coalesce)
 - IoT-focused bootstrapping (NDNViber)
- ◇ **Access control**
 - of Context-Content exchange
 - Tag-based Access Control by Routers
- ◇ **Availability**
 - Framework for DDoS Mitigation at the Edge



NDN-ABS enables **name-based schematization of signatures**
verification based on attribute-based cryptography.

Information-Centric Wireless

- ◇ **Improve AR/VR performance** beyond current TCP/IP based wireless and mobility
 - Revamp low-layer wireless for high-speed mobility support using cross-layer approaches made possible via naming
- ◇ Information-centric wireless delivery by **exploiting broadcast wireless medium**
 - Receiver-driven delivery via link-layer wireless multicast (Hybrid AP / P2P delivery in automated multi-tier/mesh multicast)
- ◇ Built-in **runtime learning/reasoning** under dynamic conditions and various failure modes
 - Devices perform local learning and reasoning, AP/NDN router makes aggregate decision. (Collect data from names, & optional interest fields.)
- ◇ Experimental results:
 - High throughput via better-utilization of the broadcast wireless media.
 - ▷ Link-layer multicast up to 450Mbps versus legacy 6Mbps.
 - ▷ Peak application speed ~100Mbps
 - Resilient multicast mechanism reduces the packet loss rate by 74.2%.
 - Extend Android Wi-Fi Direct from single-group to multi-hop multi-group.



Good fit for each other: Wireless broadcast and NDN's data-centric delivery

FPGA acceleration as a service

Pose Estimation as an Application Driver

- Application:
 - ▷ Context for AR/MR, Video surveillance, sport analysis, Driver assistance systems, etc.
- Challenges:
 - ▷ High computation cost, deep neural network
- ◇ Hardware optimizations gain on OpenPose-V2 ➡ 2.3x speedup
- ◇ Software optimizations gain on integration ➡ 5x speedup
- ◇ Overall speedup on OpenPose-V2 ➡ 11.5x speedup
 - 23.8 FPS
- ◇ Composable accelerator architecture (FlexCNN)



3.8x more energy efficient than GPU

Inference-only runtime (without integration)

CPU: 56-core Intel Xeon CPU E5-2680 v4

GPU: NVIDIA Tesla V100 using cuDNN

FPGA: Xilinx Virtex Ultrascale+ VCU1525

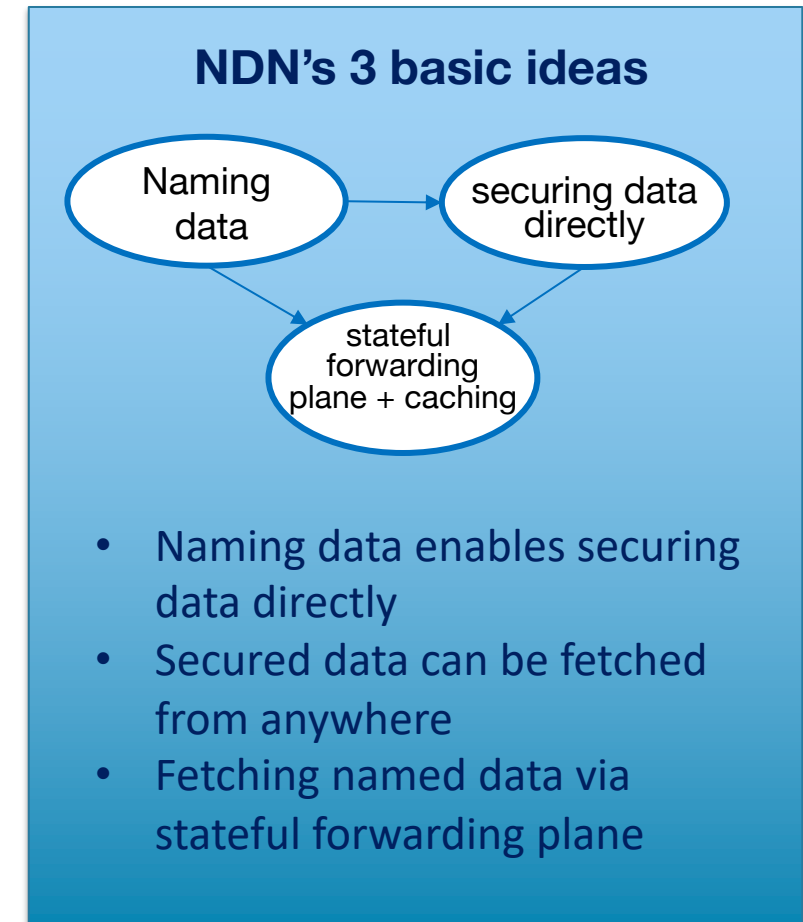
Port to Intel FPGA ongoing

Platform	Frequency (GHz)	Runtime (ms)	Dynamic Power (W)
CPU	2.4	99.3	17
GPU	1.4	25.3	38
FPGA (ours)	0.243	24.7	10

Some Contributions and Lessons Learned

Further validation of the NDN architecture design

- ◇ NDN: networking by fetching app-named, secured data
- ◇ Network, computation, and storage: all supply named data
 - more seamless integration into a cohesive, secure, and distributed ecosystem
 - novel unifications of application approaches – e.g., video as bus, stream, store
- ◇ Making app data names visible to lower layers facilitates cross-layer optimization
- ◇ Use of name semantics opens the door to fundamentally new system innovations



Deepened understanding of network architecture impact

IP's model of point-to-point packet delivery now influences everything.

- ◇ Lower-layer technology
 - e.g. commercial WiFi optimized to support IP's point-to-point delivery even over broadcast media
- ◇ Upper-layer app development
 - e.g. handling everything, including distribution, at application layer
- ◇ System platforms in the middle
 - e.g. socket API: synchronous point-to-point data delivery
- ◇ Security throughout
 - securing channels patched on: managing trust outside the app

Clearer understanding of NDN rollout challenges

Unforeseen challenges to quantitative comparisons / implementations.

- Lack of access to lower layers in protocol stack
- Media formats and protocols increasingly complex (baking in assumptions about use)
- *In situ* tests and modification to AR platforms and game engines require significant software engineering

“Why can't I do the same thing using IP?”

- You “can”...
- Telephony could also have connected all computers in the world, at that time only
- CDNs mitigated the problem of content distribution over IP infrastructure, but it too will lose effectiveness with mobiles and local/dynamic content...
- Large suite of protocols, public or semi-proprietary mitigation solutions developed

Clearer understanding of NDN rollout challenges

- ◇ **NDN**: a fundamental change to the communication model
- ◇ IP had a clean slate implementation: *adding* TCP/IP protocol stack into OS
 - Connecting IP nodes via telephone network to form IP overlay
 - Followed by clean slate app development
- ◇ OS platforms must allow deeper access to network interfaces to enable new network paradigm
- ◇ To gain the most advantage, applications over NDN should follow a data-centric paradigm
 - Some applications already do this more than others
 - Frameworks that do this at the app level probably do not do it at the network level
 - Without some awareness, hard to even gain the basics: intrinsic security, disruption tolerance, multi-interface/multi-path.

Guide future efforts toward fruitful areas: edge-in, plug-and-play, greenfields

ICE-AR Future Work

- ◇ Development of NDN APIs
 - Application use of rich NDN namespace
 - Easily leveraging security schemes
 - Supporting various pub-sub and transport functions, async/sync
 - Expansion of library-based microforwarder to support experimentation
- ◇ NDN plug-n-play: simplify platform configuration & deployment
 - particularly in supporting apps at ad hoc mobile edge with dense deployment
- ◇ Complete porting of the FlexCNN framework to Intel FPGAs
- ◇ Optimizations
 - range of implementation enhancements
 - security components (e.g., using alternative/lighter/faster signature schemes)
- ◇ Structured, qualitative evaluation of application benefits

Thank you!

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